# STACKS, DERIVED CATEGORIES OF SPACES: GENERATOR-WITH-SUPPORT

THE STACKS PROJECT AUTHORS

*Editorial note.* Statement span is lines 3996-4003 of sources/stacks-spaces-perfect/pinned-original-source.tex; proof span is lines 4005-4077, the complete author proof (71 body lines).

*Editorial note.* Navigation stubs: 8 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

*Editorial note.* extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

*Editorial note.* Text is retained verbatim, including the author's own typos and placeholders.

*Editorial note.* Queue index 24 of queue_spaceperfect.json (pb=3244, stmt 3996-4003, proof 4005-4077).

*Editorial note.* stacks-spaces-perfect wave sp01 (2026-10-14): fresh adjudication accept per lane-s3/sp01_judgments; pool per Chief 12:4x unfreeze (reserve-pool workflow, six-check clean); spaces-perfect.tex 126 with-proof rows, 70 b107-occupied labels avoided, 56 free rows judged 32 accept / 24 reject.

*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

*Source reference.* Original lemma 08DF: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 4077–4077. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 08GP: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 1563–1577. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 09IV: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 3814–3825. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 08HK: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 3610–3616. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 08GG: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 2135–2145. Cited by the retained statement or

*Date*: stacks-project:spaces-perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0A9A: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 4077–4077. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 09IR: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 4077–4077. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 08GH: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 896–962. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

**Lemma 0.1.** *Let $S$ be a scheme. Let $X$ be a quasi-compact and quasi-separated algebraic space over $S$. Let $T \subset |X|$ be a closed subset such that $|X| \setminus T$ is quasi-compact. With notation as above, the category $D_{QCoh,T}(\mathcal{O}_X)$ is generated by a single perfect object.*

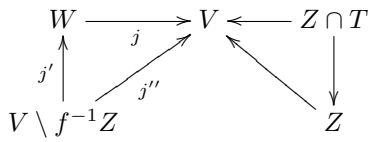
*Proof.* We will prove this using the induction principle of Lemma 08GP. The property is true for representable quasi-compact and quasi-separated objects of the site $X_{spaces,étale}$ by Derived Categories of Schemes, Lemma 0A9A.

Assume that $(U \subset X, f : V \to X)$ is an elementary distinguished square such that the lemma holds for $U$ and $V$ is affine. To finish the proof we have to show that the result holds for $X$. Let $P$ be a perfect object of $D(\mathcal{O}_U)$ supported on $T \cap U$ which is a generator for $D_{QCoh,T\cap U}(\mathcal{O}_U)$. Using Lemma 09IV we may choose a perfect object $Q$ of $D(\mathcal{O}_X)$ supported on $T$ whose restriction to $U$ is a direct sum one of whose summands is $P$. Write $V = \mathrm{Spec}(B)$. Let $Z = X \setminus U$. Then $f^{-1}Z$ is a closed subset of $V$ such that $V \setminus f^{-1}Z$ is quasi-compact. As $X$ is quasi-separated, it follows that $f^{-1}Z \cap f^{-1}T = f^{-1}(Z \cap T)$ is a closed subset of $V$ such that $W = V \setminus f^{-1}(Z \cap T)$ is quasi-compact. Thus we can choose $g_1, \ldots, g_s \in B$ such that $f^{-1}(Z \cap T) = V(g_1, \ldots, g_r)$. Let $K \in D(\mathcal{O}_V)$ be the perfect object corresponding to the Koszul complex on $g_1, \ldots, g_s$ over $B$. Note that since $K$ is supported on $f^{-1}(Z \cap T) \subset V$ closed, the pushforward $K' = R(V \to X)_*K$ is a perfect object of $D(\mathcal{O}_X)$ whose restriction to $V$ is $K$ (see Lemmas 08HK and 08GG). We claim that $Q \oplus K'$ is a generator for $D_{QCoh,T}(\mathcal{O}_X)$.

Let $E$ be an object of $D_{QCoh,T}(\mathcal{O}_X)$ such that there are no nontrivial maps from any shift of $Q \oplus K'$ into $E$. By Lemma 08GG we have $K' = R(V \to X)_!K$ and hence

$$\mathrm{Hom}_{D(\mathcal{O}_X)}(K'[n], E) = \mathrm{Hom}_{D(\mathcal{O}_V)}(K[n], E|_V)$$

Thus by Derived Categories of Schemes, Lemma 09IR we have $E|_V = Rj_*E|_W$ where $j : W \to V$ is the inclusion. Picture




Since $E$ is supported on $T$ we see that $E|_W$ is supported on $f^{-1}T \cap W = f^{-1}T \cap (V \setminus f^{-1}Z)$ which is closed in $W$. We conclude that

$$E|_V = Rj_*(E|_W) = Rj_*(Rj'_*(E|_{U\cap V})) = Rj''_*(E|_{U\cap V})$$

Here the second equality is part (1) of Cohomology, Lemma 08DF which applies because $V$ is a scheme and $E$ has quasi-coherent cohomology sheaves hence pushforward along the quasi-compact open immersion $j'$ agrees with pushforward on the underlying schemes, see Remark 08GH. This implies that $E = R(U \to X)_* E|_U$ (small detail omitted). If this is the case then

$$\mathrm{Hom}_{D(\mathcal{O}_X)}(Q[n], E) = \mathrm{Hom}_{D(\mathcal{O}_U)}(Q|_U[n], E|_U)$$

which contains $\mathrm{Hom}_{D(\mathcal{O}_U)}(P[n], E|_U)$ as a direct summand. Thus by our choice of $P$ the vanishing of these groups implies that $E|_U$ is zero. Whence $E$ is zero. $\square$